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(54) **SEMICONDUCTOR DEVICES**

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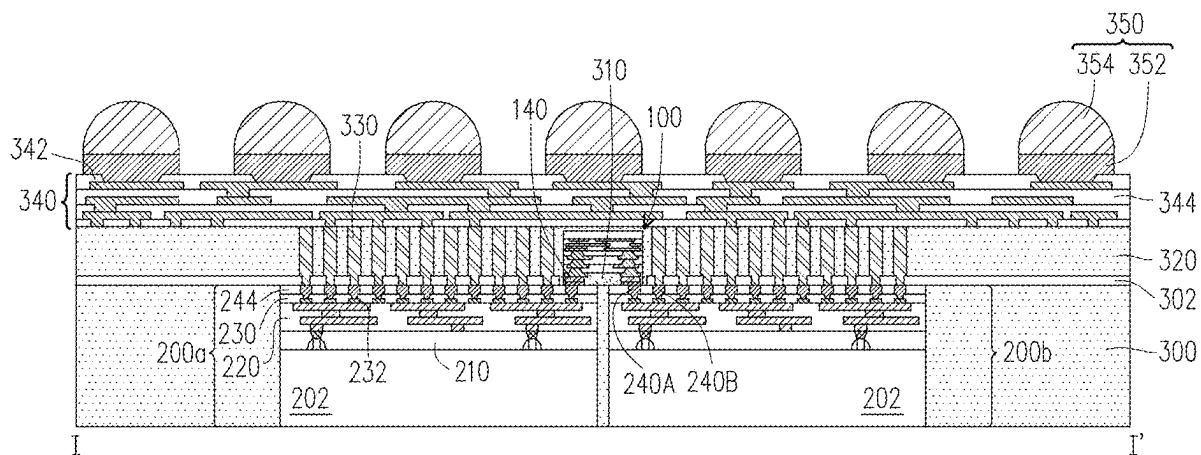
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Related U.S. Application Data

(63) Continuation of application No. 18/442,062, filed on Feb. 14, 2024, now Pat. No. 12,341,104, which is a continuation of application No. 17/884,560, filed on Aug. 9, 2022, now Pat. No. 11,935,836, which is a continuation of application No. 16/924,201, filed on Jul. 9, 2020, now Pat. No. 11,450,612.

(57) **ABSTRACT**

A semiconductor device includes a first integrated circuit. The first integrated circuit includes a first substrate, a first interconnect structure on the first substrate and a first conductive connector on the first interconnect structure. The first conductive connector includes a plurality of conductive pads and a plurality of first conductive vias and a second conductive via between the conductive pads, the second conductive via is not overlapped with the first conductive vias while the first conductive vias are overlapped with one another, and a vertical distance between the second conductive via and the first interconnect structure is smaller than a vertical distance between each of the first conductive vias and the first interconnect structure.



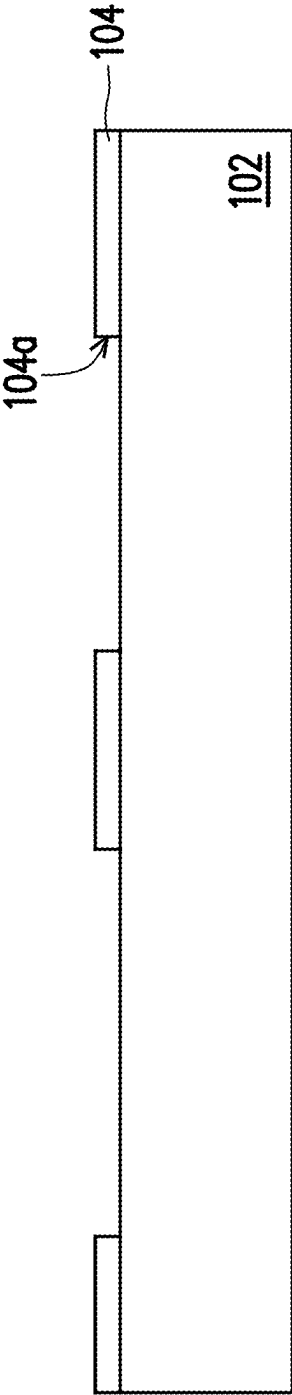


FIG. 1A

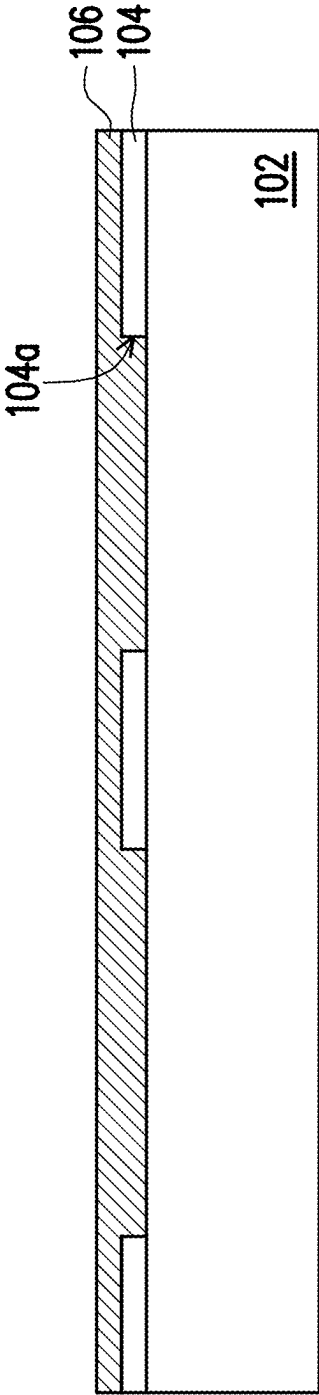


FIG. 1B

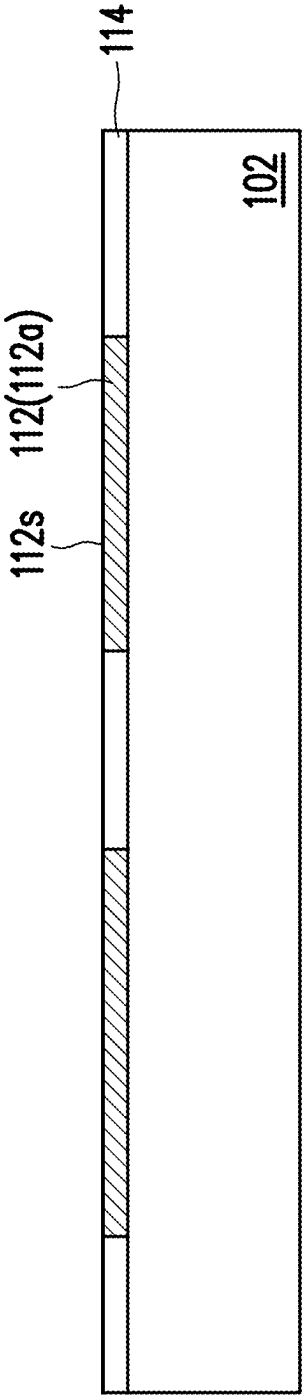


FIG. 1C

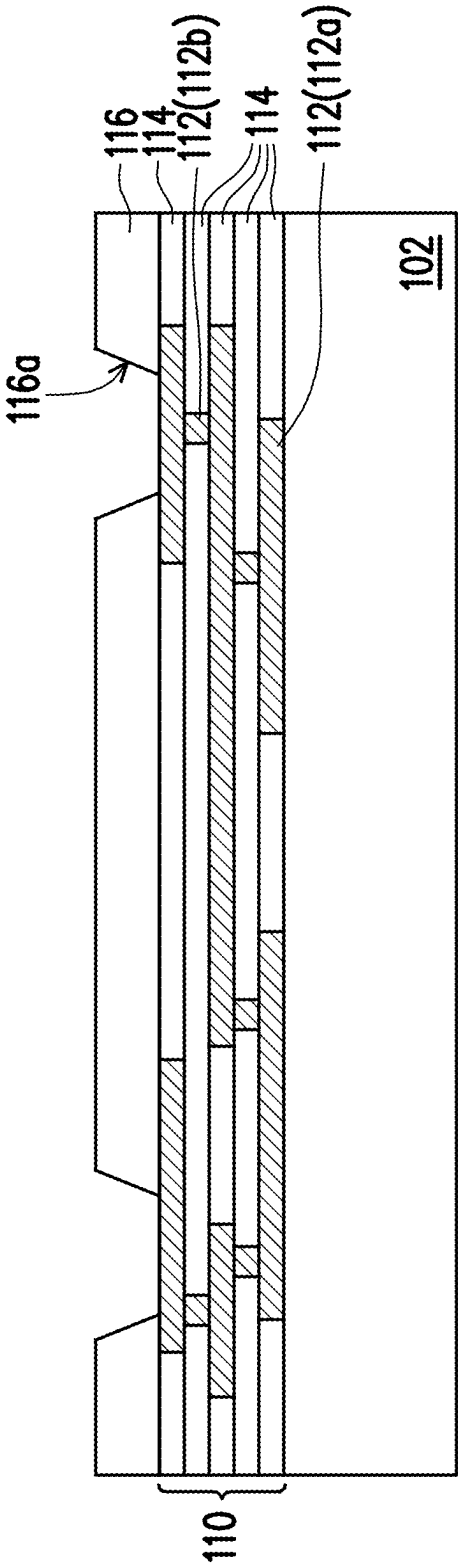


FIG. 1D

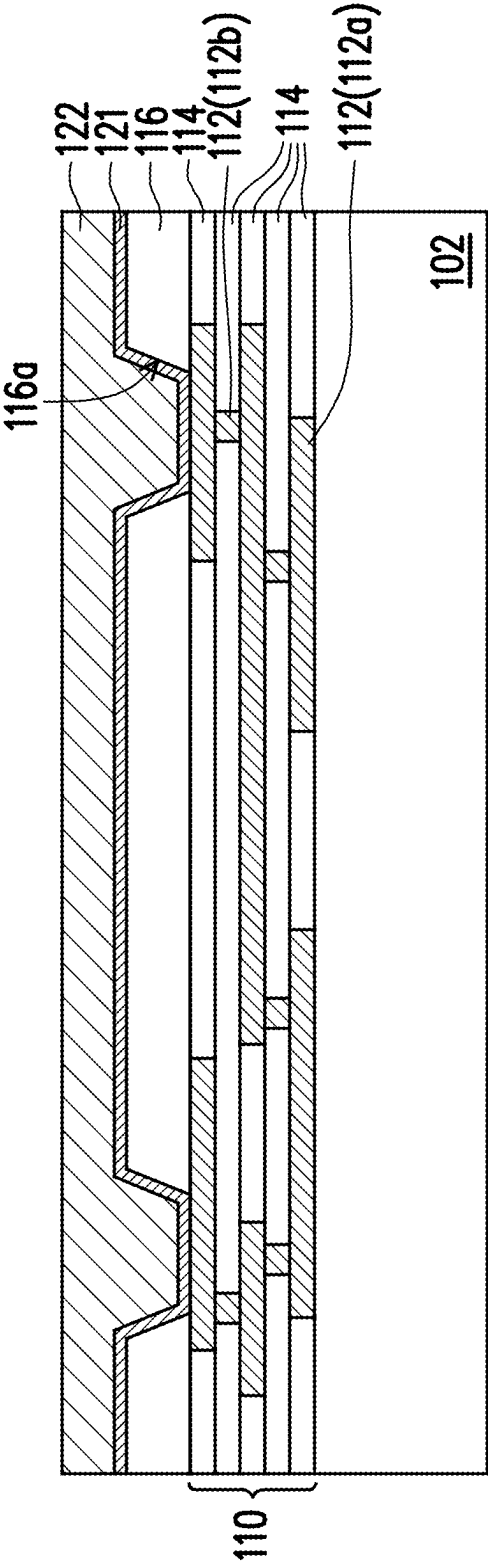


FIG. 1E

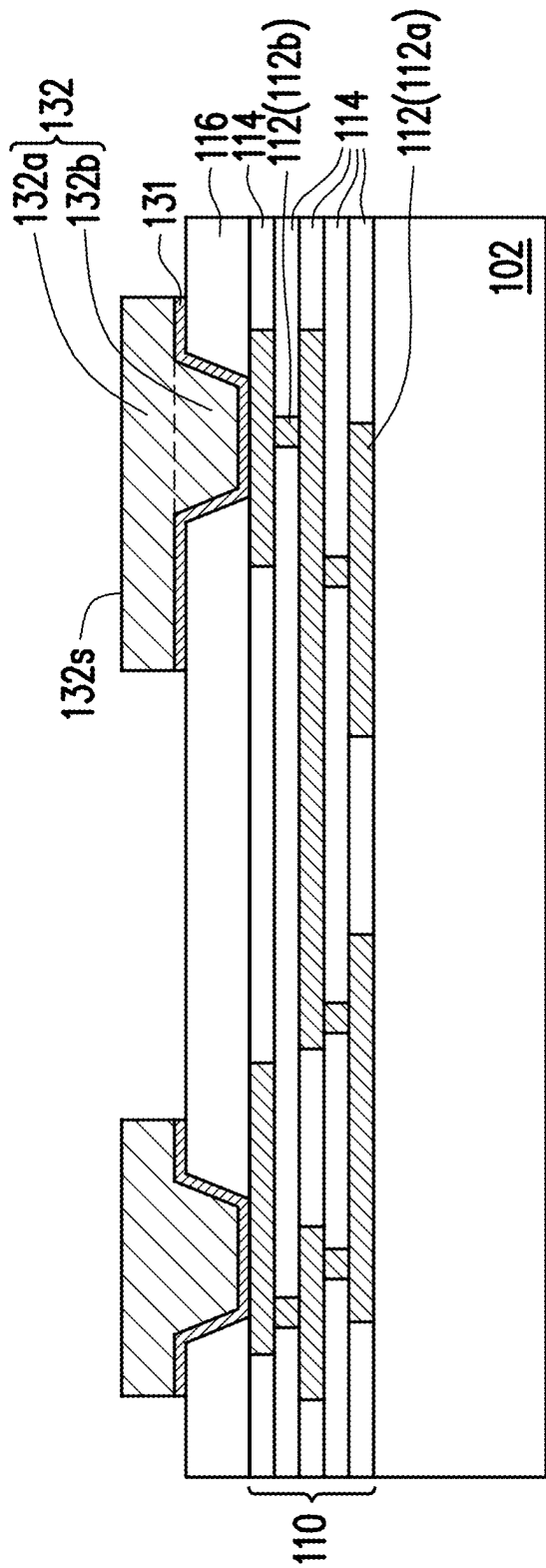


FIG. 1F

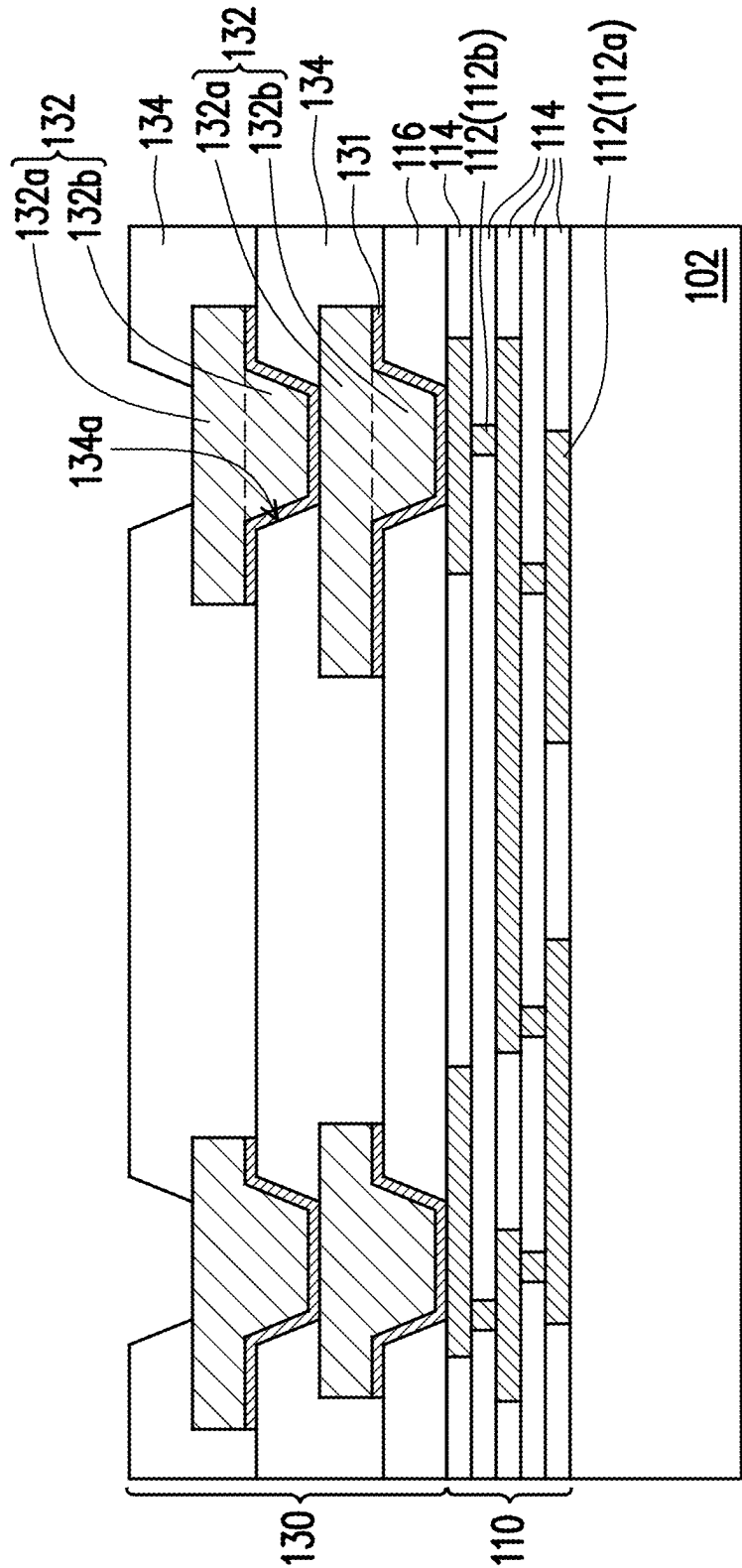


FIG. 1G

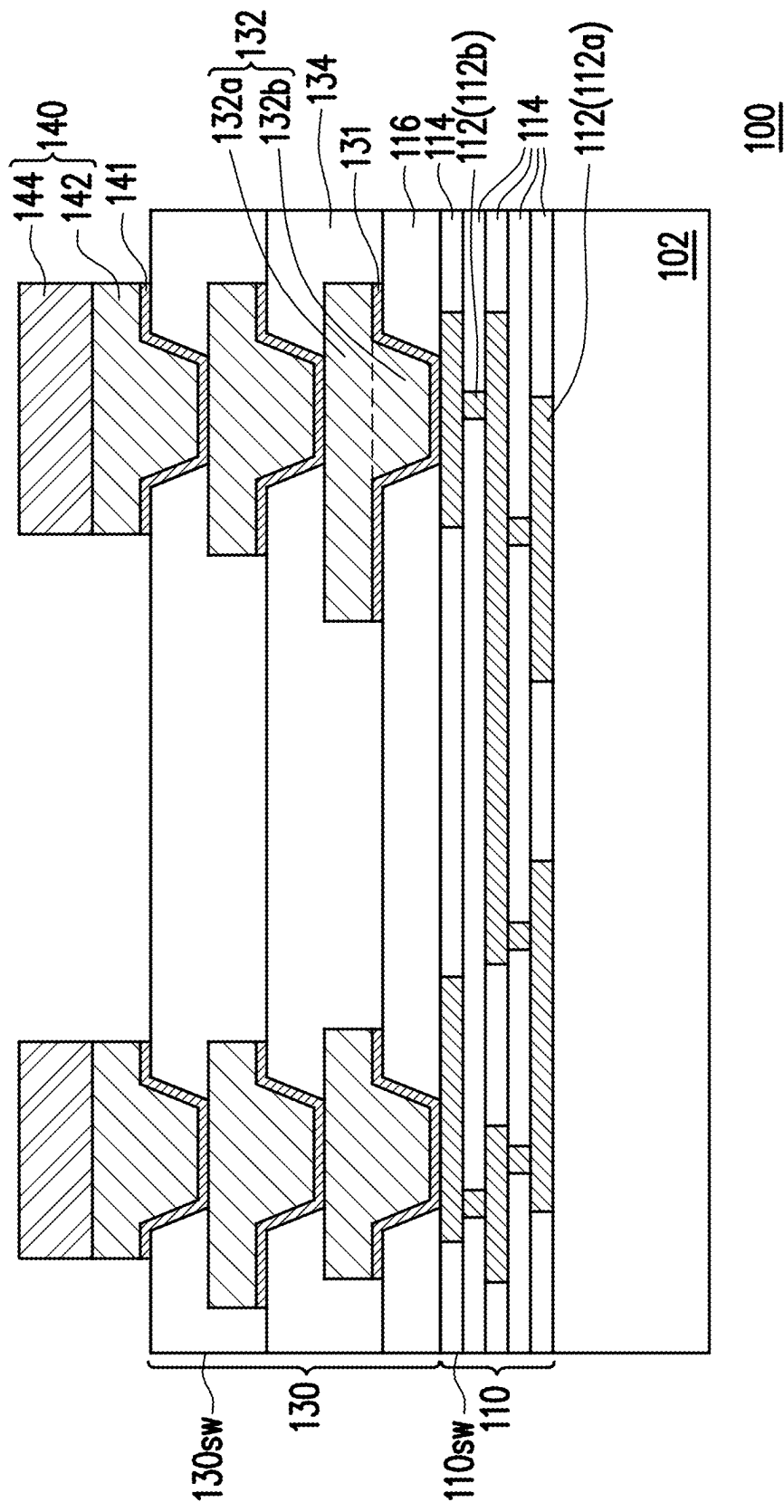


FIG. 1H

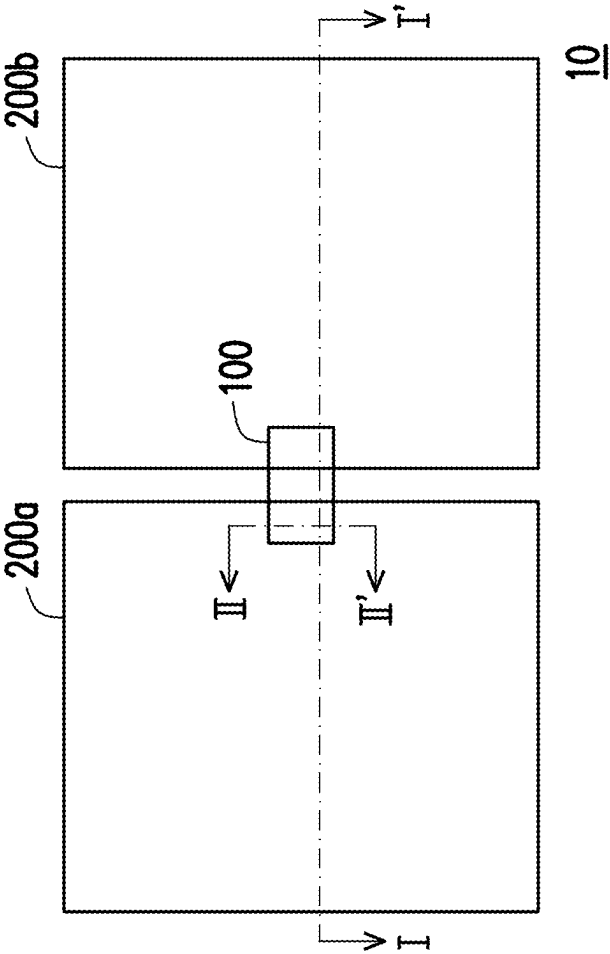


FIG. 2A

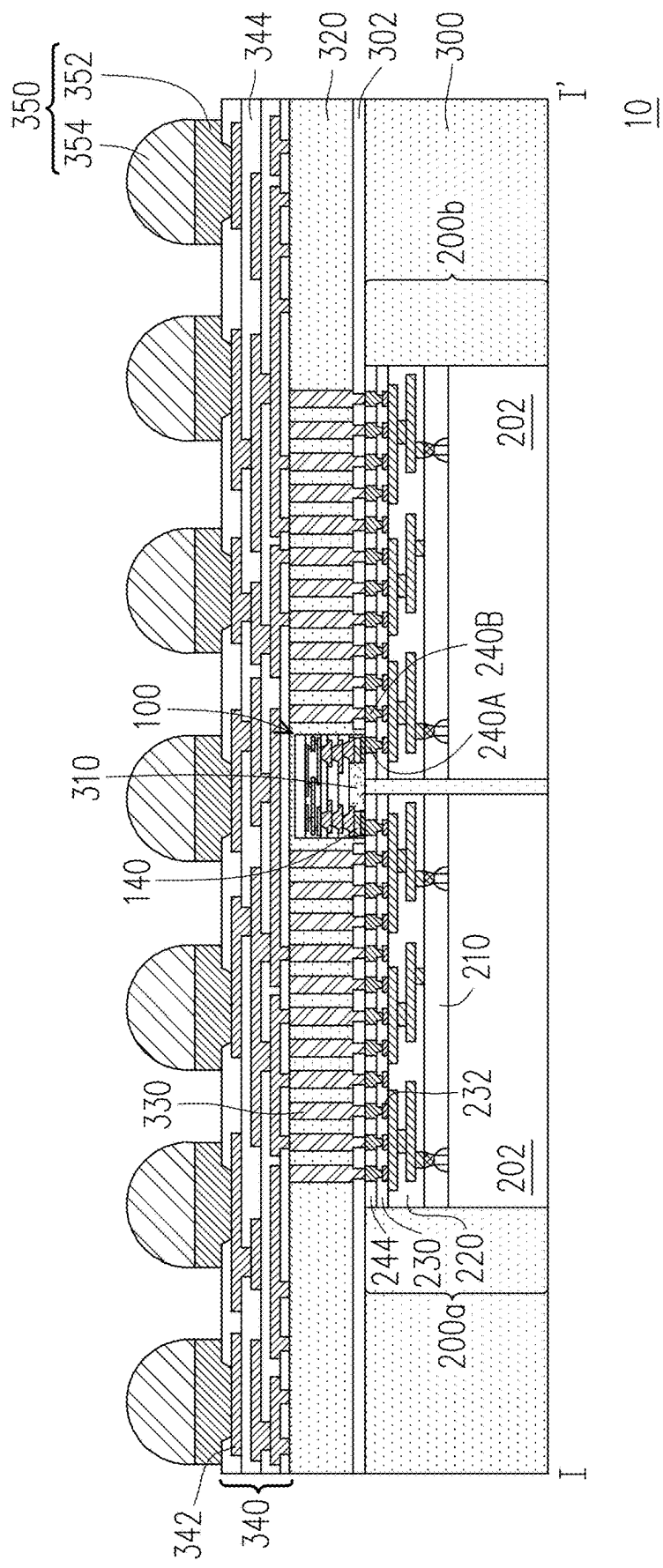


FIG. 2B

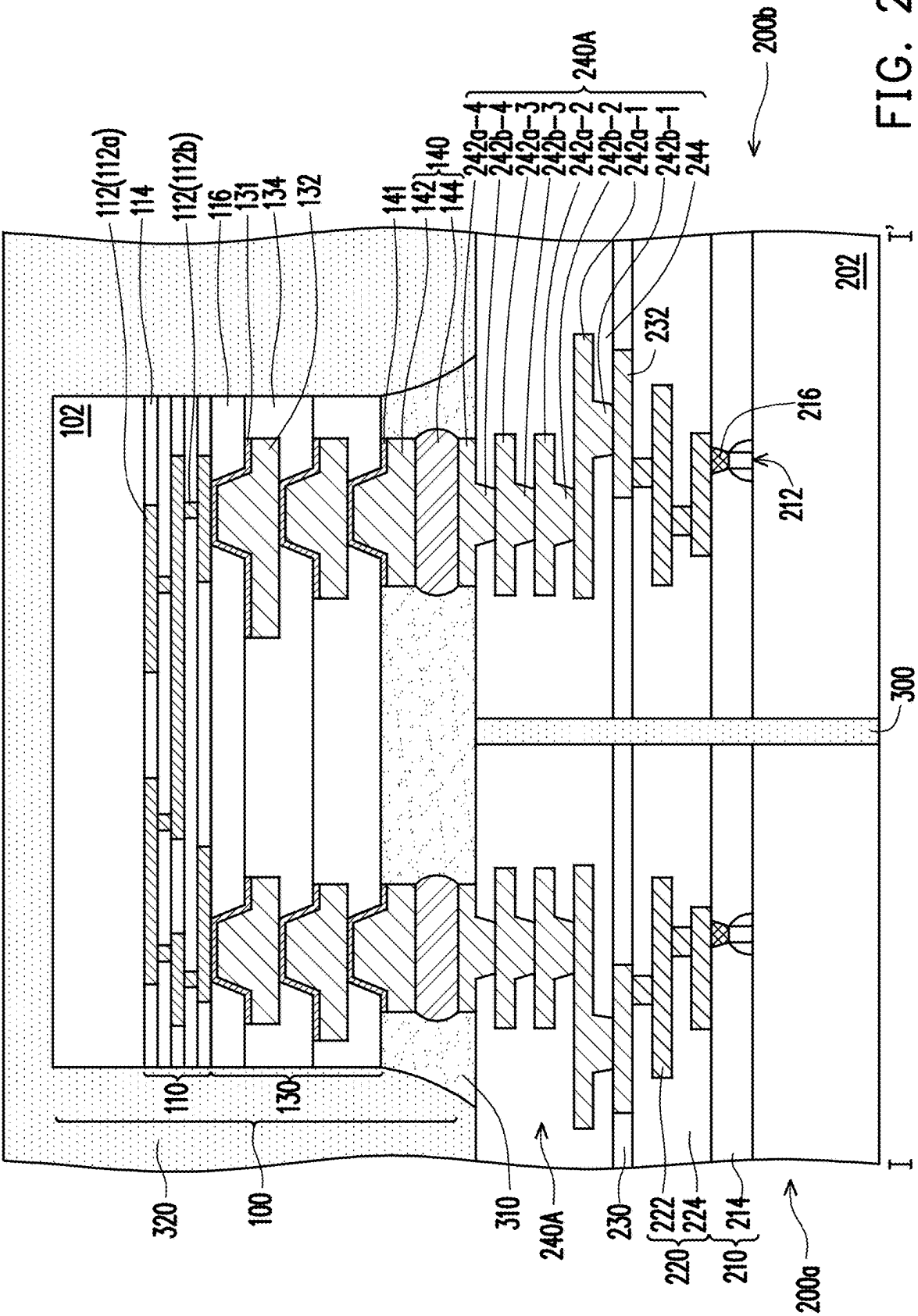


FIG. 2C

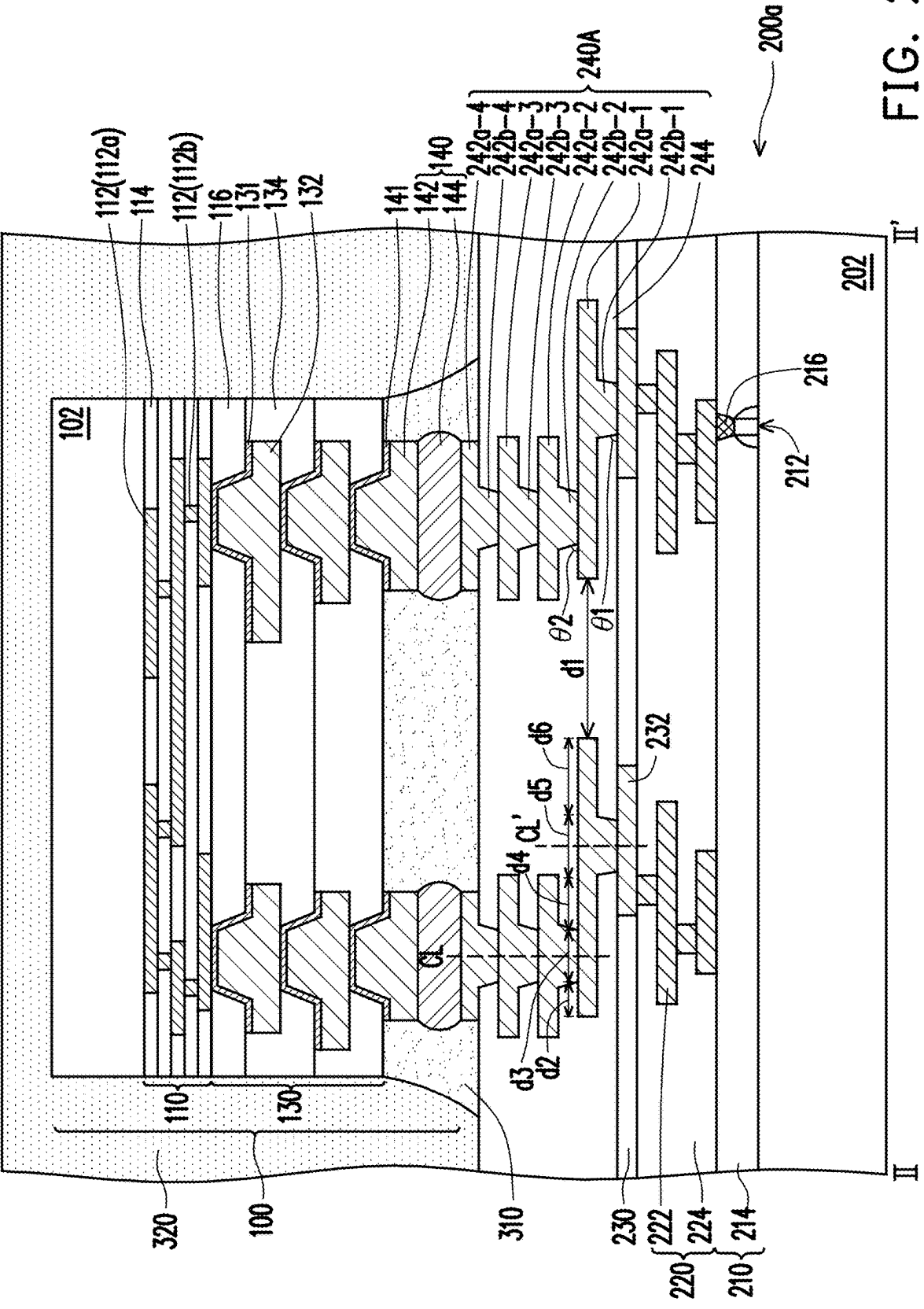


FIG. 2D

SEMICONDUCTOR DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/442,062, filed on Feb. 14, 2024, now allowed. The prior application Ser. No. 18/442,062 is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/884,560, filed on Aug. 9, 2022, now patented. The prior application Ser. No. 17/884,560 is a continuation application of and claims the priority benefit of a prior application Ser. No. 16/924,201, filed on Jul. 9, 2020, now patented. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] In recent years, the semiconductor industry has experienced rapid growth due to continuous improvement in integration density of various electronic components, e.g., transistors, diodes, resistors, capacitors, etc. For the most part, this improvement in integration density has come from successive reductions in minimum feature size, which allows more components to be integrated into a given area.

[0003] These smaller electronic components also require smaller packages that occupy less area than previous packages. Examples of the type of packages for semiconductors include quad flat pack (QFP), pin grid array (PGA), ball grid array (BGA), flip chips (FC), three-dimensional integrated circuits (3DICs), wafer level packages (WLPs), and package on package (POP) devices. Some 3DICs are prepared by placing chips over chips on a semiconductor wafer level. The 3DICs provide improved integration density and other advantages, such as faster speeds and higher bandwidth, because of the decreased length of interconnects between the stacked chips. However, there are many challenges related to 3DICs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1A to FIG. 1H are schematic cross-sectional views illustrating a method of manufacturing a bridge according to some embodiments of the disclosure.

[0006] FIG. 2A is a top view of a semiconductor device according to some embodiments of the disclosure, FIG. 2B is a schematic cross-sectional view taken along line I-I' of FIG. 2A, FIG. 2C is an enlarged view of a portion of the structure showing in FIG. 2B, and FIG. 2D is a schematic cross-sectional view taken along line II-II' of FIG. 2A.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely

examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0010] FIG. 1A to FIG. 1H are schematic cross-sectional views illustrating a method of manufacturing a bridge according to some embodiments of the disclosure.

[0011] Referring to FIG. 1A, a substrate **102** is provided. In some embodiments, the substrate **102** is a glass substrate, a ceramic substrate, a semiconductor substrate, or the like. In some embodiments, the substrate **102** is a silicon wafer or an active layer of a semiconductor-on-insulator (SOI) substrate, or the like. The substrate **102** may include a semiconductor material, such as doped or undoped silicon, or may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used.

[0012] Then, as shown in FIG. 1A to FIG. 1D, an interconnect structure **110** is formed over the substrate **102**. The interconnect structure **110** may be formed using a damascene process. In detail, as shown in FIG. 1A, first, a dielectric material **104** is formed on the substrate **102**. A material of the dielectric material **104** may include an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a nitride such as silicon nitride, or the like. In some alternative embodiments, the dielectric material **104** includes low-k dielectrics having a smaller dielectric con-

stant than silicon oxide, and examples of low-k dielectrics include organosilicate glasses (OSG) such as carbon-doped silicon dioxide and fluorine-doped silicon dioxide (otherwise referred to as fluorinated silica glass (FSG)). In some alternative embodiments, the dielectric material **104** includes extremely low-k dielectrics having a dielectric constant of about 2.1 or less, and extremely low-k dielectrics are generally low-k dielectrics formed into a porous structure. Porosity reduces the effective dielectric constant. The dielectric material **104** may be formed any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof. Then, a plurality of openings **104a** are formed in the dielectric material **104**. The openings **104a** may be formed by a photolithography process.

[0013] Referring to FIG. 1B, a conductive material **106** is formed on the dielectric material **104** and filled into the openings **104a**. A material of the conductive material **106** may include copper, silver, gold, tungsten, aluminum, copper doped with aluminum or manganese, combinations thereof, or the like. The conductive material **106** may be formed by a plating process such as electro-plating, electroless-plating, immersion plating, or the like. In some alternative embodiments, an optional diffusion barrier layer and/or optional adhesion layer may be deposited before forming the conductive material **106**. A material of the diffusion barrier layer may include titanium, titanium nitride, titanium oxide, tantalum, tantalum nitride, titanium oxide, or other alternatives, and suitable materials for the conductive material.

[0014] Referring to FIG. 1C, portions of the conductive material **106** outside the openings **104a** are removed, so as to form a plurality of conductive features **112** in a dielectric layer **114**. In some embodiments, a planarization process such as chemical mechanical planarization (CMP) process may be used to remove excess conductive material **106** from a surface of the dielectric material **104** and to planarize the surface for subsequent processing. Accordingly, the conductive features **112** may have smooth top surfaces **112s**. In some embodiments, the formed conductive features **112** are conductive lines **112a** formed by a damascene process.

[0015] Referring to FIG. 1D, a plurality of conductive features **112** a plurality of dielectric layers **114** and are formed over the bottommost conductive features **112** in the bottommost dielectric layer **114**, to form an interconnect structure **110**. The conductive features **112** may be conductive lines **112a** and/or conductive vias **112b**. The conductive vias **112b** may extend through the dielectric layer **114** to provide vertical connections between layers of conductive lines **112a**. The conductive features **112** may be formed by using a damascene process as shown in FIG. 1A and FIG. 1C, that is, forming a dielectric material, forming a plurality of openings corresponding to the desired pattern of the conductive lines and/or the conductive vias, filling the openings with a conductive material and removing excess of the conductive material outside the openings. In some alternative embodiments, the conductive features **112** may be formed by a dual damascene process, in other words, the conductive lines **112a** and the respective conductive vias **112b** are formed integrally. In some alternative embodiments, the dielectric layer **114** has a single layered or a multiple layered structure.

[0016] In some embodiments, the interconnect structure **110** may include five layers of conductive features **112** and five layers of dielectric layers **114**. In other embodiments,

the interconnect structure **110** include a different number of layers of conductive features **112** and a different number of layers of dielectric layers **114**. In some embodiments, a thickness of a layer of the conductive lines **112a** is smaller than 1 μm . In some embodiments, the use of a damascene or dual damascene process forms the conductive features **112** having a smaller linewidth, which can increase the density of the conductive features **112**. For example, the conductive features **112** may have a linewidth smaller than 1 μm .

[0017] In some embodiments, a passivation layer **116** is formed on the interconnect structure **110**. The passivation layer **116** has a plurality of openings **116a** to expose portions of the interconnect structure **110**. The passivation layer **116** may include an oxide such as silicon oxide, a nitride such as silicon nitride, or the like. The passivation layer **116** may be formed any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof.

[0018] Referring to FIG. 1E to FIG. 1G, a redistribution layer (RDL) structure **130** is formed over the interconnect structure **110**. The redistribution layer structure **130** may be formed using a photolithography process. In detail, first, as shown in FIG. 1E, a conductive material **122** is formed on the passivation layer **116** and filled into the openings **116a** of the passivation layer **116**. A material of the conductive material **122** may include copper, silver, gold, tungsten, aluminum, combinations thereof, or the like. The conductive material **122** is blanket deposited on the passivation layer **116** by sputtering, printing, electro plating, electroless plating or CVD, for example. In some embodiments, a diffusion barrier material **121** is formed between the conductive material **122** and the passivation layer **116**. The diffusion barrier material **121** may include titanium, titanium nitride, titanium oxide, tantalum, tantalum nitride, titanium oxide, or other alternatives. The diffusion barrier material **121** is blanket deposited on the passivation layer **116**, for example.

[0019] Referring to FIG. 1F, the conductive material **122** is patterned into a plurality of conductive features **132**. In some embodiments, the conductive material **122** and the diffusion barrier material **121** thereunder are patterned simultaneously by using a mask (not shown), to form the conductive features **132**. In some embodiments, the conductive features **132** are disposed on the passivation layer **116** and fills the openings **116a**, so as to electrically connect to the conductive features **112** of the interconnect structure **110**. In some embodiments, each of the conductive features **132** include a conductive line **132a** and a conductive via **132b**. A diffusion barrier layer **131** is disposed under a bottom surface of the conductive feature **132**, for example. In some embodiments, a sidewall of the diffusion barrier layer **131** is substantially flush with a sidewall of the conductive feature **132**. In some embodiments, since the conductive features **132** are formed by the photolithography process, the conductive features **132** may have rough top surfaces **132s**. The roughness of the top surfaces **132s** of the conductive features **132** may be larger than the roughness of the top surfaces **122s** of the conductive features **122**.

[0020] Referring to FIG. 1G, at least one polymer layer **134** and a plurality of conductive features **132** are formed over the bottommost conductive features **132**, to form a RDL structure **130**. The conductive features **132** may be formed by using a photolithography process, that is, forming the polymer layer **134** including a plurality of openings **134a** to expose the conductive features **132**, forming a conductive

material over the polymer layer **134**, and patterning the conductive material. A material of the polymer layer **134** may include organic materials such as polyimide, polybenzoxazole (PBO), benzocyclobutene (BCB), or the like. The polymer layers **134** may be formed by any acceptable deposition process, such as spin coating, CVD, laminating, the like, or a combination thereof. The material of the polymer layer **134** is different from the material of the dielectric layer **114**. In some embodiments, the material of the polymer layer **134** is different from the material of the passivation layer **116**. A material of the conductive material may include copper, silver, gold, tungsten, aluminum, combinations thereof, or the like. The conductive material is deposited on the polymer layer **134** by sputtering, printing, electro plating, electroless plating or CVD, for example. In some embodiments, the conductive line **132a** and the conductive via **132b** are integrally formed. However, the disclosure is not limited thereto. In some alternative embodiments, the conductive line **132a** and the conductive via **132b** are separately formed, and the conductive line **132a** and the conductive via **132b** are disposed in different polymer layers **134**. In some alternative embodiments, the polymer layer **134** has a single layered or a multiple layered structure.

[0021] In some embodiments, the RDL structure **130** includes the conductive features **132** (sometimes referred to as redistribution layers or redistribution lines) in the polymer layers **134**. The conductive features **132** may be conductive lines **132a** or conductive vias **132b** between the conductive lines **132a**. The conductive vias **132b** may extend through the polymer layer **134** to provide vertical connections between layers of conductive lines **132a**. Some of the conductive vias **132b** may extend through the passivation layer **116** to provide vertical connections between the RDL structure **130** and the interconnect structure **110**. The RDL structure **130** may include four stacked layers of conductive features and two layers of polymer layers. In other embodiments, the RDL structure **130** includes a different number of layers of conductive features and a different number of layers of polymer layers. For example, the RDL structure **130** includes at least one layer of conductive feature and at least one layer of polymer layer. In some embodiments, a thickness of a layer of the conductive lines **132a** is larger than 4 μm .

[0022] Referring to FIG. 1H, a plurality of conductive connectors **140** are formed over the RDL structure **130**. Thus, a bridge **100** is completed. In some embodiments, the conductive connectors **140** are referred to as conductive terminals. In some embodiments, the conductive connectors **140** includes conductive pads or conductive pillars **142** with solder regions **144** disposed thereon. In some embodiments, the conductive pillars **142** may have substantially vertical sidewalls. In some alternative embodiments, the conductive pillars **142** may be omitted, and the solder regions **144** may be disposed directly on the conductive features **132** of the RDL structure **130**. In some embodiments, the conductive pillars **142** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **140** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **140** may be formed by a suitable process such as

evaporation, plating, ball drop, screen printing, or a ball mounting process. In some embodiments, a diffusion barrier layer **141** is disposed under a bottom surface of the conductive pillar **142**. A material of the diffusion barrier layer **141** may include titanium, titanium nitride, titanium oxide, tantalum, tantalum nitride, titanium oxide, or other alternatives. In some embodiments, the conductive connectors **140** are arranged along a periphery of the bridge **100**. For example, the conductive connectors **140** are disposed at four sides of the bridge **100**. In some alternative embodiments, the conductive connectors **140** are disposed at opposite sides of the bridge **100**. In some embodiments, two conductive connectors **140** are disposed at one side of the bridge **100**. However, the disclosure is not limited thereto. The bridge **100** may have any number of conductive connectors **140** at one side. In some alternative embodiments, multiple bridges are formed on a single substrate (i.e., the substrate **102**) and singulated in to form individual bridge, such as the individual interconnect device **100** shown in FIG. 1H. In such embodiments, after forming the conductive connectors **140**, a dicing process is performed.

[0023] The bridge **100** has the substrate **102**, the interconnect structure **110**, the RDL structure **130** and the conductive connectors **140**. In some embodiments, the bridge **100** is free of active devices. For example, the bridge **100** may be free of transistors, diodes, and/or the like. Further, the bridge **100** may be also be free of passive devices such as capacitors, resistors, inductors, and/or the like. The RDL structure **130** is disposed between the interconnect structure **110** and the conductive connectors **140**. The conductive connectors **140** are electrically connected to each other through the interconnect structure **110** and the RDL structure **130**. In some embodiments, a sidewall **110_{sw}** of the interconnect structure **110** (i.e., sidewalls of the dielectric layers **114**) is substantially flush with a sidewall **130_{sw}** of the RDL structure **130** (i.e., sidewalls of the polymer layers **134**). The bridge **100** provides electrical connection between devices directly bonded to the conductive connectors **140** such as between devices dies (for example, a logic die and a memory die). In embodiments where the substrate **102** includes silicon, the bridge **100** is also referred to as a silicon bus or a silicon bridge.

[0024] FIG. 2A is a top view of a semiconductor device according to some embodiments of the disclosure, FIG. 2B is a schematic cross-sectional view taken along line I-I' of FIG. 2A, FIG. 2C is an enlarged view of a portion of the structure showing in FIG. 2B, and FIG. 2D is a schematic cross-sectional view taken along line II-II' of FIG. 2A. For clarity, details of bridge and integrated circuits are omitted in FIG. 2B.

[0025] Referring to FIG. 2A to FIG. 2D, a semiconductor device **10** includes a plurality of integrated circuits **200** and a bridge **100** electrically connecting the integrated circuits **200**. In some embodiments, the bridge **100** has a structure the same as or similar to the bridge **100** of FIG. 1H. In some embodiments, the integrated circuits **200** are disposed side by side and encapsulates by an encapsulant **300**. The integrated circuits **200** may be of different types or the same type. For example, the integrated circuit **200** may be a logic die (e.g., central processing unit (CPU), graphics processing unit (GPU), system-on-a-chip (SoC), application processor (AP), microcontroller, etc.), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management die (e.g.,

power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies), the like, a multi-functional die, or combinations thereof. The integrated circuits 200 have similar structures. In some embodiments, widths and lengths of the integrated circuits 200 are different, and heights of the integrated circuits 200 are substantially the same. In some alternative embodiments, widths, lengths and/or heights of the integrated circuits 200 are substantially the same or different.

[0026] Referring to FIG. 2C and FIG. 2D, in some embodiments, the integrated circuit 200 includes a substrate 202, a device layer 210 in and/or on the substrate 202, an interconnect structure 220 over the device layer 210, a passivation layer 230 over the interconnect structure 220, a plurality of conductive pads 232 in the passivation layer 230 and a plurality of conductive connectors 240A, 240B in a dielectric layer 244 over the conductive pads 232. In some embodiments, the substrate 202 is similar to the substrate 102. The device layer 210 is disposed on a top surface of the substrate 202 or partially disposed in the substrate 202. The device layer 210 includes at least one device 212 in a dielectric layer 214, for example. The device 212 may be active devices such as transistors. The device 212 is electrically connected to the interconnect structure 220 through a via 216. The interconnect structure 220 may include a plurality of conductive features 222 in a dielectric layer 224 to provide circuit structure. The conductive features 222 may include a plurality of conductive lines and a plurality of conductive vias interconnecting the conductive lines. The conductive pads 232 are electrically connected to the interconnect structure 220. The conductive connectors 240A, 240B are electrically connected to the conductive pads 232 respectively.

[0027] Referring to FIG. 2B to FIG. 2D, in some embodiments, sidewalls of the integrated circuits 200 are encapsulated by the encapsulant 300, and a portion of the encapsulant 300 fills a gap between the integrated circuits 200. In some embodiments, the encapsulant 300 includes a molding compound, a molding underfill, a resin such as epoxy, a combination thereof, or the like. In some other embodiments, the encapsulant 300 includes a photo-sensitive material such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), a combination thereof, or the like, which may be easily patterned by exposure and development processes or laser drilling process. In some alternative embodiments, the encapsulant 300 includes nitride such as silicon nitride, oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a combination thereof, or the like. In some embodiments, the encapsulant 300 is formed by forming an encapsulant material by a suitable fabrication technique such as spin-coating, lamination, deposition, or similar processes. The encapsulant material encapsulates top surfaces and sidewalls of the integrated circuits 200 and fills the gap between the integrated circuits 200. Thereafter, a grinding or polishing process is performed to remove a portion of the encapsulant material, such that the top surfaces of the integrated circuits 200 are exposed. In some embodiments, the top surfaces of the integrated circuits 200 and the top surface of the encapsulant 300 are substantially coplanar. In some alternative embodiments, a dielectric layer

302 is formed on the integrated circuits 200 and the encapsulant 300. The dielectric layer 302 includes a plurality of openings to expose the conductive connectors 240A, 240B of the integrated circuits 200.

[0028] The bridge 100 is bonded to the integrated circuits 200 through the conductive connectors 140. For example, the solder regions 144 of the conductive connector 140 are bonded to the conductive connectors 240A of the integrated circuits 200 using a flip chip bonding process. A reflow process may be applied to adhere the solder regions 144 of the conductive connectors 140 to the conductive connectors 240A. In some embodiments, after the reflow process, the solder region 144 of the conductive connector 140 is rounded. Conventionally, during the joint process of the bridge and the integrated circuits under a temperature change from a low temperature such as 100° C. to a high temperature such as 300° C., the warpage profile of the bridge (there is no RDL structure in conventional bridge), the warpage profile of the bridge remains “smiling” (concave) warpage profile. On contrary, in some embodiments, since the bridge 100 is configured to have the RDL structure 130 including polymer material, the warpage profile of the bridge 100 changes from a “crying” (convex) warpage profile to a “smiling” (concave) warpage profile during the joint process of the bridge 100 and the integrated circuits 200 under a temperature change from a low temperature such as 100° C. to a high temperature such as 300° C. Thus, the joint window may be improved.

[0029] In some embodiments, after the reflow process, an underfill layer 310 is formed to fill the space between the bridge 100 and the integrated circuits 200. The underfill layer 310 covers a first surface (i.e., a surface facing the integrated circuits 200) of the bridge 100, sidewalls of the conductive connectors 140, a surface of the dielectric layer 244 and sidewalls of the conductive connectors 240A. In some alternative embodiments, the underfill layer 310 further extends upward to cover a portion of a sidewall of the bridge 100. In some embodiments, the underfill layer 310 may be a polymer such as epoxy.

[0030] In some embodiments, after forming the underfill layer 310, an encapsulant 320 is formed over the integrated circuits 200 to encapsulate the sidewall of the bridge 100 and a sidewall of the underfill layer 310. The encapsulant 320 is disposed over the integrated circuits 200 and the encapsulant 300. In some embodiments, a second surface opposite to the first surface of the bridge 100 is covered by the encapsulant 320. In some alternative embodiments, the second surface of the bridge 100 is coplanar with a surface of the encapsulant 320. In some embodiments, an outer sidewall of the encapsulant 320 is substantially flush with an outer sidewall of the encapsulant 300. In some embodiments, the forming method and the material of the encapsulant 320 are similar to those of the encapsulant 300.

[0031] In some embodiments, a plurality of through integrated fan-out vias (TIVs) 330 are formed in the encapsulant 320 to electrically connect to the integrated circuits 200. In some embodiments, the TIVs 330 penetrate the encapsulant 320 and the dielectric layer 302. The TIVs 330 may be in electrical contact with the conductive connectors 240B (which are not bonded to the bridge 100) of the integrated circuits 200. The TIVs 330 may surround the bridge 100. The TIVs 330 include conductive materials such as copper,

nickel, solder, alloys thereof, or the like, or a combination thereof. The TIVs 330 are formed by, electroplating, for example.

[0032] In some embodiments, after forming the encapsulant 320 and the TIVs 330, a RDL structure 340 is formed over the encapsulant 320, the TIVs 330 and the bridge 100. The RDL structure 340 is electrically connected to the TIVs 330, such that the two integrated circuits 200 are electrically connected through the TIVs 330 and the RDL structure 340. In some embodiments, the RDL structure 340 includes a plurality of conductive features 342 and a plurality of polymer layers 344. The conductive features 342 are stacked alternately in the polymer layers 344. In some embodiments, the forming method and the material of the RDL structure 340 are similar to those of the RDL structure 130. In some embodiments, an outer sidewall of the RDL structure 340 are substantially flush with the outer sidewall of the encapsulant 300 and the outer sidewall of the encapsulant 320.

[0033] In some embodiments, a plurality of conductive connectors 350 are formed over the RDL structure 340. In some embodiments, the conductive connectors 350 are referred to as conductive terminals. The conductive connectors 350 may include conductive pads or conductive pillars 352 with solder regions 354 disposed thereon. In some embodiments, the conductive connectors 350 are similar to the conductive connectors 140. In some embodiments, a diameter of the conductive connectors 350 is larger than a diameter of the conductive connectors 140. The conductive connectors 350 are electrically connected to the two integrated circuits 200 through the RDL structure 340 and the TIVs 330. In some embodiments, the two integrated circuits 200 are electrically connected through the bridge 100, the TIVs 330 and the RDL structure 340. In some alternative embodiment, the semiconductor device 10 is placed on a package structure such as an integrated circuit (IC) device to form a POP device. The POP device may be further connected to other package components such as a printed circuit board (PCB), a flex PCB, or the like.

[0034] In some embodiments, as shown in FIG. 2C and FIG. 2D, the conductive connector 240A include a plurality of conductive pads 242a-1, 242a-2, 242a-3, 242a-4 and a plurality of conductive vias 242b-1, 242b-2, 242b-3, 242b-4 stacked on one another. The conductive vias 242b-1, 242b-2, 242b-3, 242b-4 are disposed between the conductive pads 232, 242a-1, 242a-2, 242a-3, 242a-4. The conductive pad 242a-1, 242a-2, 242a-3, 242a-4 and the respective conductive via 242b-1, 242b-2, 242b-3, 242b-4 may be integrally formed. In some alternative embodiments, the conductive pad 242a-1, 242a-2, 242a-3, 242a-4 and the respective conductive via 242b-1, 242b-2, 242b-3, 242b-4 are separated formed. In some embodiments, the conductive via 242b-1 is directly connected to the conductive pad 232 in the passivation layer 230. In some embodiments, the integrated circuits 200 are each directly bonded to the conductive connectors 140. That is, the conductive pad 242a-4 may be directly bonded to the conductive connector 140 of the bridge 100. For example, the conductive pads 242a-4 are in direct contact with the solder regions 144 of the conductive connector 140.

[0035] In some embodiments, at the location directly below the conductive connectors 140 of the bridge 100, the conductive vias 242b-1, 242b-2, 242b-3, 242b-4 of the conductive connectors 240A are stacked sequentially over the substrate 202. That is, a distance (i.e., a vertical distance)

between the conductive via 242b-1 and the conductive connector 140 of the bridge 100 is larger than a distance between the conductive via 242b-2, 242b-3, 242b-4 and the conductive connector 140 of the bridge 100. In some embodiments, the conductive vias 242b-2, 242b-3, 242b-4 are substantially aligned with one another while the conductive via 242b-1 is misaligned with the conductive vias 242b-2, 242b-3, 242b-4. For example, central lines CL of the conductive vias 242b-2, 242b-3, 242b-4 are substantially aligned, and a central line CL' of the conductive via 242b-1 is not separated from and not aligned with the central lines CL of the conductive vias 242b-2, 242b-3, 242b-4. In some embodiments, the conductive vias 242b-2, 242b-3, 242b-4 are partially overlapped in a vertical direction, and the conductive via 242b-1 is not overlapped with the conductive vias 242b-2, 242b-3, 242b-4 in the vertical direction. That is, the conductive via 242b-1 is not directly stacked with the conductive vias 242b-2, 242b-3, 242b-4. In some embodiments, stacking conductive vias 242b-2, 242b-3, 242b-4 prevent cracks in the fine pitch structure, and the conductive via 242b-1 misaligned with the stacking conductive vias 242b-2, 242b-3, 242b-4 prevents high stress induced void. In an embodiment, compared to the configuration of the conductive vias 242b-1, 242b-2, 242b-3, 242b-4 being aligned with one another, the configuration of the conductive vias 242b-1, 242b-2, 242b-3, 242b-4 of FIG. 2C and FIG. 2D prevents the crack. For example, the occurrence of the crack is reduced to about 0% from about 38%. In some alternative embodiments, the conductive via 242b-1, 242b-2, 242b-3 are substantially aligned with one another while the conductive via 242b-4 is misaligned with them.

[0036] In an embodiment, a material of the dielectric layer 244 surrounding the conductive via 242b-1 is epoxy. The critical dimension of the conductive via 242b-1, 242b-2 may be smaller than the critical dimension of the conductive via 242b-3, 242b-4. In an embodiment, the critical dimension of the conductive via 242b-1 is substantially the same as the critical dimension of the conductive via 242b-2. In an embodiment, the conductive via 242b-3 is large than the critical dimension of the conductive via 242b-1, 242b-2, 242b-4. In some embodiments, the conductive connectors 240A of the integrated circuits 200 directly bonded to the bridge 100 have the configuration of FIG. 2C and FIG. 2D. However, the disclosure is not limited thereto. In some alternative embodiments, the conductive connector 240A has other suitable configuration.

[0037] Conventionally, the bridge (i.e., bridge die) is fabricated merely by using damascene process. On contrary, in some embodiments, the bridge is fabricated by using both damascene process and RDL process. In other words, the bridge is configured to have the RDL structure including polymer material. Accordingly, the warpage profile of the bridge changes from a “crying” (convex) warpage profile to a “smiling” (concave) warpage profile during the joint process of the bridge and the integrated circuits under a temperature change from a low temperature to a high temperature. Thus, the joint window may be improved. Additionally, in some embodiments, by stacking conductive vias in a specific arrangement, cracks in the fine pitch structure and high stress induced void are prevented.

[0038] In accordance with some embodiments of the disclosure, a semiconductor device includes a bridge and a plurality of dies. The bridge is free of active devices and includes a substrate, an interconnect structure, a RDL struc-

ture and a plurality of conductive connectors. The interconnect structure includes at least one dielectric layer and a plurality of first conductive features in the at least one dielectric layer. The RDL structure includes at least one polymer layer and a plurality of second conductive features in the at least one polymer layer, wherein a sidewall of the interconnect structure is substantially flush with a sidewall of the RDL structure. The conductive connectors are electrically connected to one another by the RDL structure and the interconnect structure. The bridge electrically connects the plurality of dies.

[0039] In accordance with some embodiments of the disclosure, a semiconductor device includes a bridge, a first integrated circuit and a second integrated circuit. The bridge is free of active devices includes a semiconductor substrate, an interconnect structure, a RDL structure and a plurality of first conductive connectors. The RDL structure is disposed between the interconnect structure and the plurality of first conductive connectors. The first integrated circuit and the second integrated circuit are electrically connected by the bridge. The first integrated circuit includes a second conductive connector directly bonded to one of the first conductive connectors. The second conductive connector includes a plurality of conductive pads and a plurality of conductive vias between the plurality of conductive pads. One of the plurality of conductive vias is not overlapped with others of the plurality of the conductive vias while the others of the plurality of the conductive vias are overlapped with one another.

[0040] In accordance with some embodiments of the disclosure, a method of manufacturing a semiconductor device includes forming a bridge free of active devices, and bonding the bridge onto a first integrated circuit and a second integrated circuit to electrically connect a first integrated circuit and a second integrated circuit. A forming method of the bridge includes the following steps. A plurality of first conductive features are formed in a plurality of dielectric layers on a substrate by a damascene process, to form an interconnect structure. A plurality of second conductive features are formed in at least one polymer layer on the plurality of dielectric layers by a lithography process, to form a RDL structure over the interconnect structure. A plurality of conductive connectors are formed to electrically connect to the plurality of second conductive features.

[0041] In accordance with some embodiments of the disclosure, a semiconductor device includes a bridge and a first integrated circuit. The bridge is free of active devices and includes a first conductive connector. The first integrated circuit includes a substrate and a second conductive connector disposed in a first dielectric layer over the substrate. The second conductive connector is directly bonded to the first conductive connector. The second conductive connector includes a plurality of conductive pads and a plurality of first conductive vias and a second conductive via between the conductive pads. The second conductive via is not overlapped with the first conductive vias while the first conductive vias are overlapped with one another. A vertical distance between the second conductive via and the first conductive connector is larger than a vertical distance between each of the first conductive vias and the first conductive connector, and a sidewall of the first dielectric layer is substantially flush with a sidewall of the substrate.

[0042] In accordance with some embodiments of the disclosure, a semiconductor device includes a bridge, a first integrated circuit, a second integrated circuit, a first encapsulant, a second encapsulant and a redistribution layer structure. The bridge is free of active devices. The first integrated circuit and the second integrated circuit are electrically connected by the bridge. The first encapsulant encapsulates the first integrated circuit and the second integrated circuit. The second encapsulant encapsulates the bridge. The redistribution layer structure is disposed over the first encapsulant and the second encapsulant, and a sidewall of the redistribution layer structure is substantially flush with sidewalls of the first encapsulant and the second encapsulant.

[0043] In accordance with some embodiments of the disclosure, a semiconductor device includes a bridge and a plurality of dies. The bridge is free of active devices. The bridge includes a germanium semiconductor substrate, an interconnect structure, a redistribution layer structure and a plurality of conductive connectors. The interconnect structure includes at least one first dielectric layer and a plurality of first conductive features in the at least one first dielectric layer. The redistribution layer structure includes at least one second dielectric layer and a plurality of second conductive features in the at least one second dielectric layer, and a sidewall of the interconnect structure is substantially flush with a sidewall of the redistribution layer structure. The conductive connectors are electrically connected to one another by the redistribution layer structure and the interconnect structure. The dies are electrically connected by the bridge.

[0044] In accordance with some embodiments of the disclosure, a method of manufacturing a semiconductor device includes the following steps. A semiconductor substrate is provided. A plurality of dielectric layers and a plurality of first conductive features in the dielectric layers are formed on the semiconductor substrate. At least one polymer layer and a plurality of second conductive features in the at least one polymer layer on the dielectric layers are formed. A plurality of conductive connectors are formed to electrically connect to the second conductive features. The semiconductor substrate, the dielectric layers and the at least one polymer layer are cut into a plurality of dies.

[0045] In accordance with some embodiments of the disclosure, a method of manufacturing a semiconductor device includes the following steps. A first integrated circuit and a second integrated circuit are provided, wherein the first integrated circuit includes a first conductive via and a plurality of first conductive pads and a plurality of second conductive vias alternately stacked on the first conductive via, and the second integrated circuit includes a third conductive via and a plurality of second conductive pads and a plurality of fourth conductive vias alternately stacked on the third conductive via. A bridge die is bonded to the first integrated circuit and the second integrated circuit through the outermost first conductive pad and the outermost second conductive pad, wherein a horizontal distance between the first conductive via and the second conductive via is larger than a horizontal distance between two adjacent of the third and fourth conductive vias.

[0046] In accordance with some embodiments of the disclosure, a method of manufacturing a semiconductor device includes the following steps. An interconnect structure is formed on a substrate. A first conductive via is formed on the interconnect structure. A plurality of conductive pads and a

plurality of second conductive vias are alternately formed on the first conductive via, wherein the second conductive vias are overlapped with one another and are not overlapped with the first conductive via.

[0047] In accordance with some embodiments of the disclosure, a semiconductor device includes a first integrated circuit. The first integrated circuit includes a first substrate, a first interconnect structure on the first substrate and a first conductive connector on the first interconnect structure. The first conductive connector includes a plurality of conductive pads and a plurality of first conductive vias and a second conductive via between the conductive pads, the second conductive via is not overlapped with the first conductive vias while the first conductive vias are overlapped with one another, and a vertical distance between the second conductive via and the first interconnect structure is smaller than a vertical distance between each of the first conductive vias and the first interconnect structure.

[0048] In accordance with some embodiments of the disclosure, a semiconductor device includes a first die and a second die. The first die includes a first substrate and a first conductive terminal, the first conductive terminal includes a plurality of first conductive pads and a plurality of first conductive vias alternately stacked on a second conductive via, wherein the first conductive vias are aligned with one another, and a vertical distance between the second conductive via and the first substrate is smaller than a vertical distance between each of the first conductive vias and the first substrate. The second die includes a second substrate and a second conductive terminal, the second conductive terminal includes a plurality of second conductive pads and a plurality of third conductive vias alternately stacked on a fourth conductive via, wherein the third conductive vias are aligned with one another, a vertical distance between the fourth conductive via and the second substrate is smaller than a vertical distance between each of the third conductive vias and the second substrate, and a horizontal distance between the second conductive via and the fourth conductive via is larger than a horizontal distance between two adjacent of the first and third conductive vias.

[0049] In accordance with some embodiments of the disclosure, a semiconductor device includes a first die and a second die. The first die includes a first substrate, a first conductive terminal and a second conductive terminal. The first conductive terminal includes a plurality of first conductive pads and a plurality of first conductive vias alternately stacked on a second conductive via, wherein the first conductive vias are overlapped with one another, and a vertical distance between the second conductive via and the first substrate is smaller than a vertical distance between each of the first conductive vias and the first substrate. The second conductive terminal includes a plurality of second conductive pads and a plurality of third conductive vias alternately stacked on a fourth conductive via, wherein the third conductive vias are aligned with one another, and a vertical distance between the fourth conductive via and the first substrate is smaller than a vertical distance between each of the third conductive vias and the first substrate. The second die includes a third conductive terminal and a fourth conductive terminal. The third conductive terminal is bonded to the first conductive terminal and includes a plurality of fifth conductive vias, wherein the fifth conductive vias are overlapped with the first conductive vias and not overlapped with the second conductive via. The fourth conductive terminal is

bonded to the second conductive terminal and includes a plurality of sixth conductive vias, wherein the sixth conductive vias are overlapped with the third conductive vias and not overlapped with the fourth conductive via.

[0050] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:
 - a first integrated circuit, comprising a first substrate, a first interconnect structure on the first substrate and a first conductive connector on the first interconnect structure, wherein the first conductive connector comprises a plurality of conductive pads and a plurality of first conductive vias and a second conductive via between the conductive pads, the second conductive via is not overlapped with the first conductive vias while the first conductive vias are overlapped with one another, and a vertical distance between the second conductive via and the first interconnect structure is smaller than a vertical distance between each of the first conductive vias and the first interconnect structure.
2. The semiconductor device according to claim 1, further comprising a first dielectric layer encapsulating the first conductive connector, wherein a sidewall of the first dielectric layer is substantially flush with sidewalls of the first interconnect structure and the first substrate.
3. The semiconductor device according to claim 2, wherein a horizontal distance between the second conductive via and the sidewall of the first dielectric layer is larger than a horizontal distance between the first conductive vias and the sidewall of the first dielectric layer.
4. The semiconductor device according to claim 2, further comprising a first encapsulant encapsulating the first substrate, the first interconnect structure and the first dielectric layer.
5. The semiconductor device according to claim 1, further comprising a second integrated circuit comprising a second conductive connector bonded to the first conductive connector of the first integrated circuit.
6. The semiconductor device according to claim 5, further comprising an underfill surrounding the outermost conductive pad of the first conductive connector and the second conductive connector.
7. A semiconductor device, comprising:
 - a first die, comprising a first substrate and a first conductive terminal, the first conductive terminal comprising a plurality of first conductive pads and a plurality of first conductive vias alternately stacked on a second conductive via, wherein the first conductive vias are aligned with one another, and a vertical distance between the second conductive via and the first substrate is smaller than a vertical distance between each of the first conductive vias and the first substrate; and

a second die, comprising a second substrate and a second conductive terminal, the second conductive terminal comprising a plurality of second conductive pads and a plurality of third conductive vias alternately stacked on a fourth conductive via, wherein the third conductive vias are aligned with one another, a vertical distance between the fourth conductive via and the second substrate is smaller than a vertical distance between each of the third conductive vias and the second substrate, and a horizontal distance between the second conductive via and the fourth conductive via is larger than a horizontal distance between two adjacent of the first and third conductive vias.

8. The semiconductor device according to claim 7, wherein central lines of the first conductive vias are aligned with one another.

9. The semiconductor device according to claim 7, wherein central lines of the third conductive vias are aligned with one another.

10. The semiconductor device according to claim 7, wherein the first die is electrically connected to the second die through electrically connecting the first conductive terminal and the second conductive terminal.

11. The semiconductor device according to claim 7, further comprising a first encapsulant encapsulating the first die and the second die.

12. The semiconductor device according to claim 11, further comprising a redistribution layer structure over the first die, the second die and the first encapsulant, wherein a sidewall of the redistribution layer structure is substantially flush with a sidewall of the first encapsulant.

13. The semiconductor device according to claim 12, further comprising a third die bonded to the first conductive terminal and the second conductive terminal and disposed between the redistribution layer structure and the first die and between the redistribution layer structure and the second die.

14. The semiconductor device according to claim 13, further comprising a second encapsulant encapsulating the third die, wherein a sidewall of the second encapsulant is substantially flush with sidewalls of the redistribution layer structure and the first encapsulant.

15. The semiconductor device according to claim 7, further comprising an underfill surrounding the outermost first conductive pad of the first conductive terminal and the outermost second conductive pad of the second conductive terminal.

16. A semiconductor device, comprising:

a first die, comprising:

a first substrate;

a first conductive terminal, comprising a plurality of first conductive pads and a plurality of first conductive vias alternately stacked on a second conductive via, wherein the first conductive vias are overlapped with one another, and a vertical distance between the second conductive via and the first substrate is smaller than a vertical distance between each of the first conductive vias and the first substrate; and

a second conductive terminal, comprising a plurality of second conductive pads and a plurality of third conductive vias alternately stacked on a fourth conductive via, wherein the third conductive vias are aligned with one another, and a vertical distance between the fourth conductive via and the first substrate is smaller than a vertical distance between each of the third conductive vias and the first substrate; and

a second die, comprising:

a third conductive terminal bonded to the first conductive terminal, comprising a plurality of fifth conductive vias, wherein the fifth conductive vias are overlapped with the first conductive vias and not overlapped with the second conductive via; and

a fourth conductive terminal bonded to the second conductive terminal, comprising a plurality of sixth conductive vias, wherein the sixth conductive vias are overlapped with the third conductive vias and not overlapped with the fourth conductive via.

17. The semiconductor device according to claim 16, wherein the first die has a first side and a second side opposite to the first side, the second conductive via is offset from the first conductive vias along a direction from the first side to the second side, and the fourth conductive via is offset from the third conductive vias along the direction from the first side to the second side.

18. The semiconductor device according to claim 16, wherein the second die further comprises a second substrate and a plurality of polymer layers encapsulating the fifth conductive vias and the sixth conductive vias, and a thickness of the polymer layers increases as the polymer layers become closer to the first die.

19. The semiconductor device according to claim 16, further comprising a first solder region between and in direct contact with the first conductive terminal and the third conductive terminal.

20. The semiconductor device according to claim 16, wherein the second conductive via is horizontally disposed between the first conductive vias and the third conductive vias and horizontally disposed between the fifth conductive vias and the sixth conductive vias.

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